

EN06 – RUST for safety critical systems

In this podcast episode, **Florian Gilcher**, Managing Director of Ferrous Systems and co-founder of the Rust Foundation, explores the role of the **Rust programming language** in mission and safety-critical systems.

Summary

Defining Mission and Safety-Critical Systems

Gilcher distinguishes between two types of high-stakes systems:

- **Mission-critical:** Systems where failure results in **fiscal or operational harm**, such as a major web service or data center outage.
- **Safety-critical:** Systems where failure puts **human lives at stake** or causes physical harm, such as in automotive, avionics, or medical devices.

The sources highlight that the line between these two is increasingly **blurred** because modern life depends on digital infrastructure like data centers, meaning a mission-critical failure can have **safety-critical repercussions** down the line.

Technical Advantages of Rust

Rust is identified as a suitable tool for these systems due to its **memory safety and concurrency protection**. Key technical features discussed include:

- **Ownership and Borrowing:** The compiler tracks data as it enters and exits the program to ensure every reference is legal, preventing issues like null pointer dereferences.
- **No Runtime:** Unlike languages with garbage collectors, Rust executes directly on the processor with **runtime components equivalent to C**, making it highly traceable and fit for operating systems or microcontrollers.
- **Compile-time Enforcement:** Patterns like **RAII** (Resource Acquisition Is Initialization) are enforced by the compiler, allowing developers to catch bugs early in the process.

Standards and Compiler Qualification

Safety-critical industries are guided by strict engineering standards such as **ISO 26262** (Automotive), **IEC 61508** (Industrial), and **DO-178** (Avionics). These standards often "highly recommend" using **statically typed languages** and restricted pointer usage—criteria that Rust meets by design.

Hiring and the Future of Rust

Gilcher notes that the **Rust job market is maturing**, evidenced by employee mobility and specialized recruiters. He advises that hiring should focus on **experienced systems engineers**, as roughly **60% to 70% of their expertise** (hardware knowledge and system architecture) is directly transferable to Rust.